AWS Revision

aws training and certification

1

aws training and certification

Introduction to Cloud

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2

AWS regions

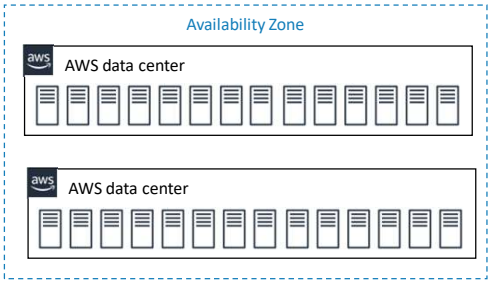
- An AWS Region is a geographical area in the world where AWS has multiple data centers.
- 39 geographic regions around the world.



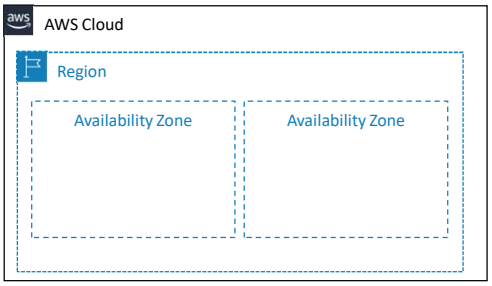
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Regions and Availability Zones



- One or more data centers
- Designed for fault isolation
- Interconnected with other Availability Zones using high-speed private links



- Geographic locations
- Two or more Availability Zones

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AWS Local Zones

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- AWS Local Zone is an AWS infrastructure location closer to end users than the main AWS Region.
- It provides very low-latency access to AWS services for users in distant cities or locations.



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AWS Wavelength Zones

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- AWS Wavelength Zone brings AWS services closer to 5G mobile users by placing AWS infrastructure within a telecom provider's 5G network.
- It provides ultra-low latency for applications such as gaming, IoT and real-time applications.



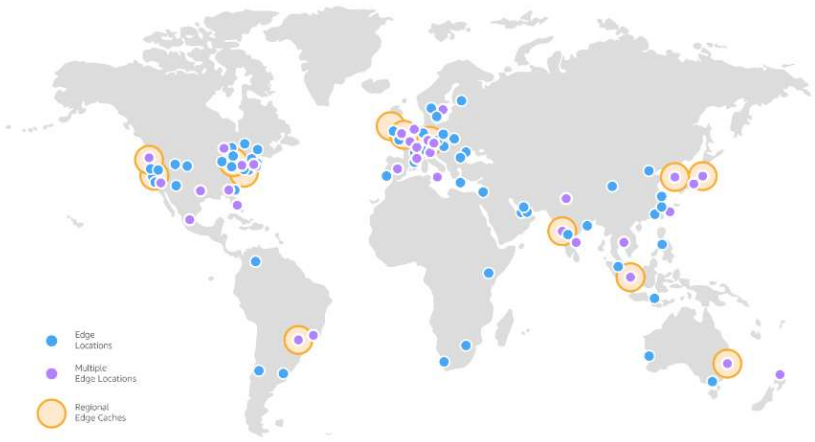
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Edge locations

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- AWS Edge Location is a location where AWS places servers close to end users to deliver content and services faster.
- It is mainly used by Amazon CloudFront to cache and deliver content with low latency.



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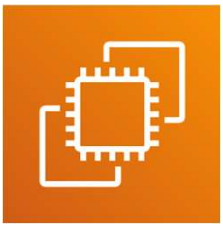
Amazon Elastic Compute Cloud (Amazon EC2)

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Amazon EC2

- Use secure, sizable compute capacity
- Boot server instances in minutes
- Pay only for what you use
- Launch an instance:
  - Select a Region
  - Select Availability Zones



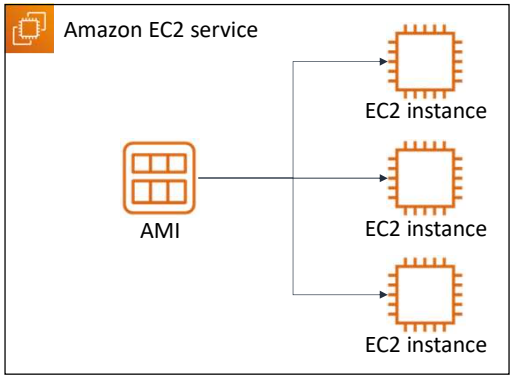
Amazon EC2

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Amazon Machine Images (AMI)

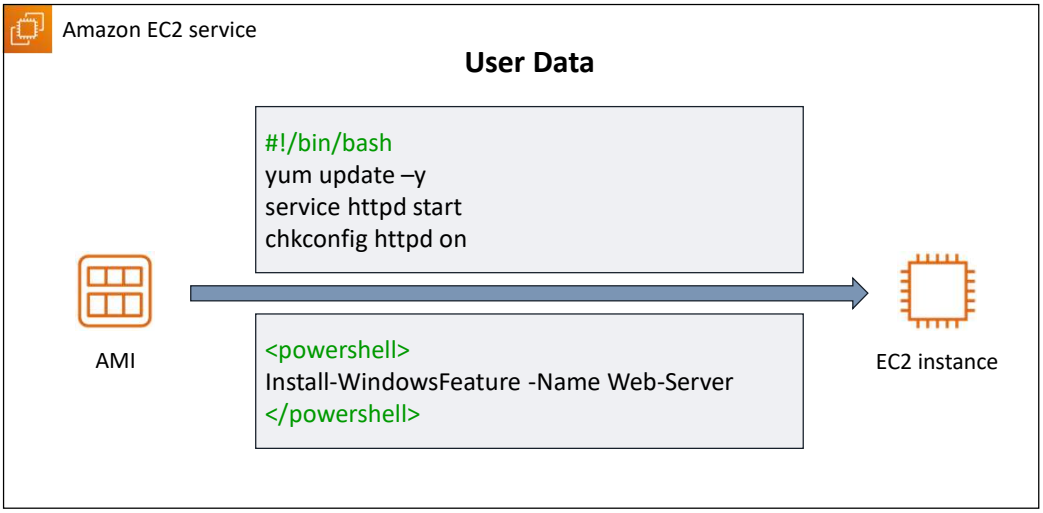
- AMI components
  - Template for instance volumes
  - Launch permissions
  - Block device mappings
- AMI sources
  - AWS
  - User-built
  - AWS user community
  - AWS Marketplace



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User data



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Amazon EC2 instance pricing options

On-Demand

- No upfront costs or minimum contracts
- Ideal for short-term, irregular workloads

Spot

- Offers savings over On-Demand prices
- Provides a 2-minute interruption notice
- Ideal for workloads with flexible start and end times

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Amazon EC2 instance pricing options

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Reserved

- Provides a billing discount over On-Demand pricing
- Requires a 1-year or 3-year term commitment

Compute Savings Plans

- Offers savings for a consistent amount of compute usage by committing to a specific hourly spend rather than a specific EC2 instance.
- Require a 1-year or 3-year term commitment

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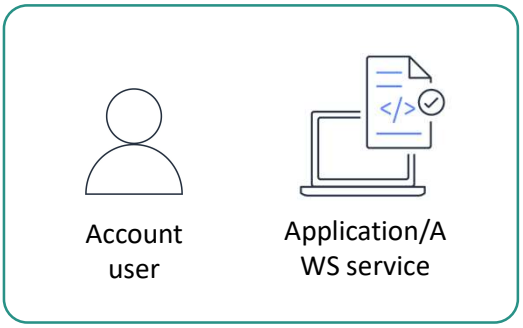
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IAM

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IAM users



An IAM user is an identity with:

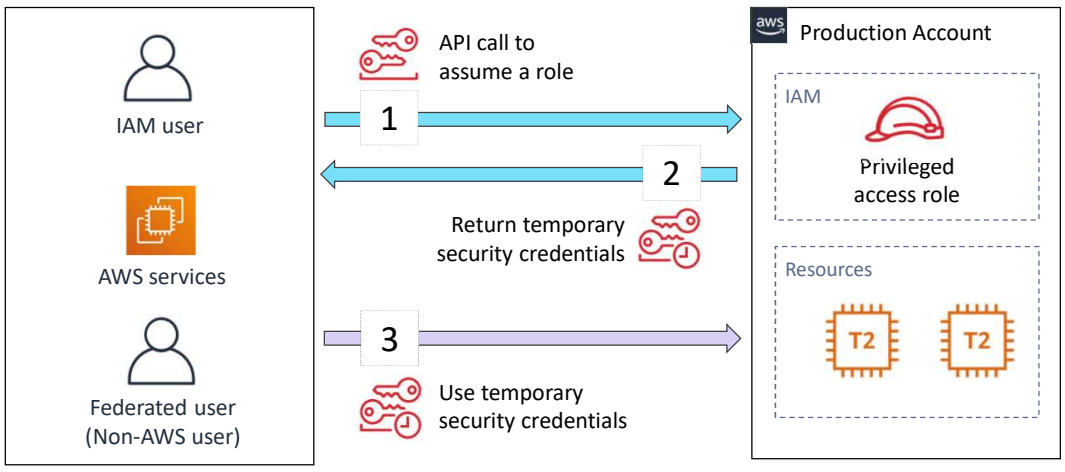
- User name and password
- Access keys
- Permissions

Best practice: Create IAM users with required permissions instead of using the root account.

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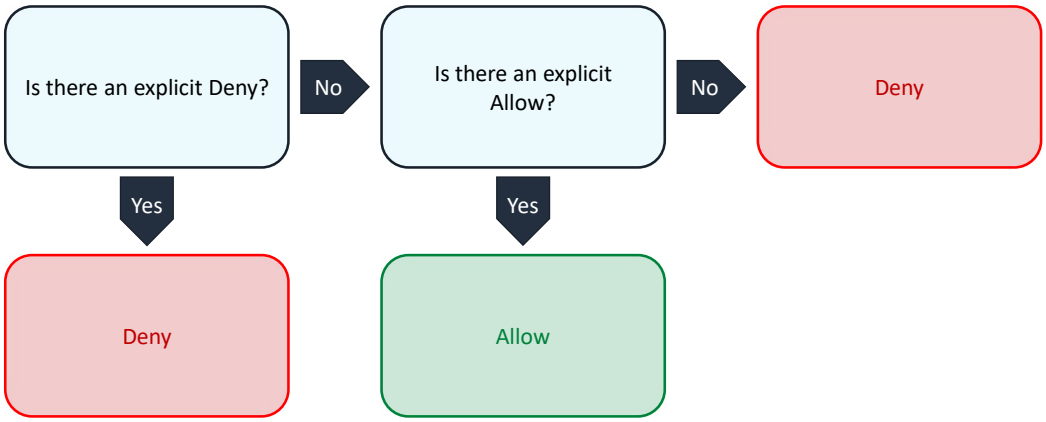
IAM roles



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Policy precedence



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Tags

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What are tags?

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Assign metadata to your resources to help with the following tasks:

- Manage
- Search
- Filter



Key	Value
example-corp:owner	John Stiles
example-corp:cost-center	78214
example-corp:environment-type	Development



Key	Value
example-corp:owner	Jane Doe
example-corp:cost-center	21002
example-corp:environment-type	Production

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Resource groups

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Resource groups



- Create, view, or manage logical groups of resources.
- Update and manage:
  - Specific applications
  - Specific layers of an application stack
  - Production versus development environments
  - Perform bulk actions on these resources

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AWS System manager

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Patch Manager

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Patch Manager

Deploy OS and software patches automatically across large groups of EC2 instances or on-premises instances.

1. 
   Create a patch baseline.
2. 
   Create a maintenance window for patching.
3. 
   Apply patches.
4. 
   Audit results with patch compliance.

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Maintenance Windows

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Maintenance Windows

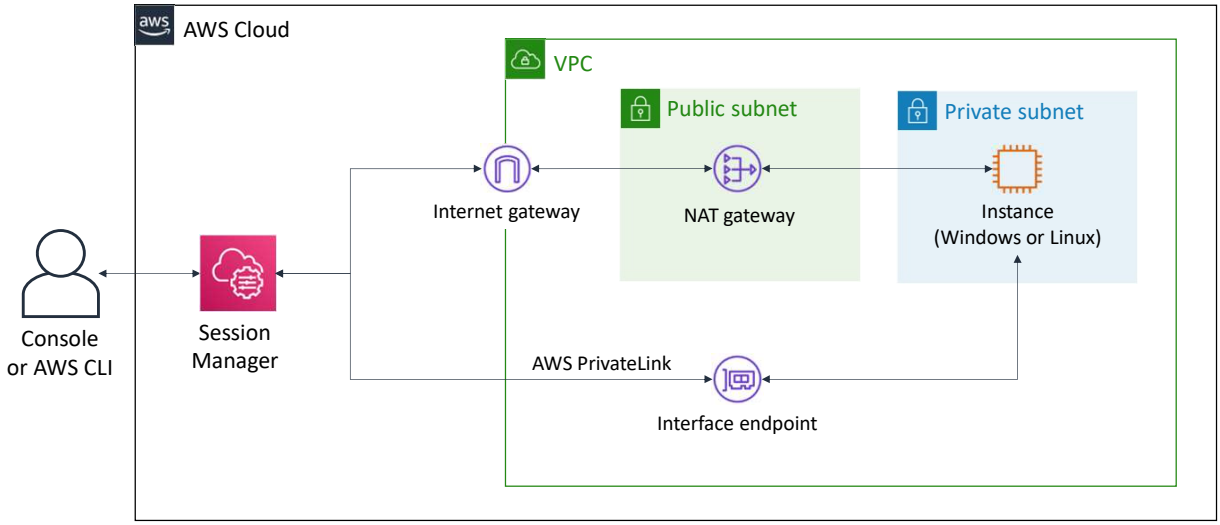
Schedule windows of time to run administrative and maintenance tasks across your instances.

1. 
   Create a maintenance window.
2. 
   Assign targets.
3. 
   Assign tasks.
4. 
   Review status after task is complete.

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Session Manager



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Automation

Automation

Safely automate common and repetitive IT operations and management tasks across AWS resources.

1. Create an Automation document.
2. Run the Automation document.
3. Monitor and test Automation.

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AWS Config

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AWS Config: Use cases

Change Management



- Mapping of resource relationships (dependencies)
- Configuration history
- Information to assess the impact of a change



AWS Config

Compliance



- Multi-account, multi-Region aggregated data
- Compliance status of resource
- Centralized console to manage resources

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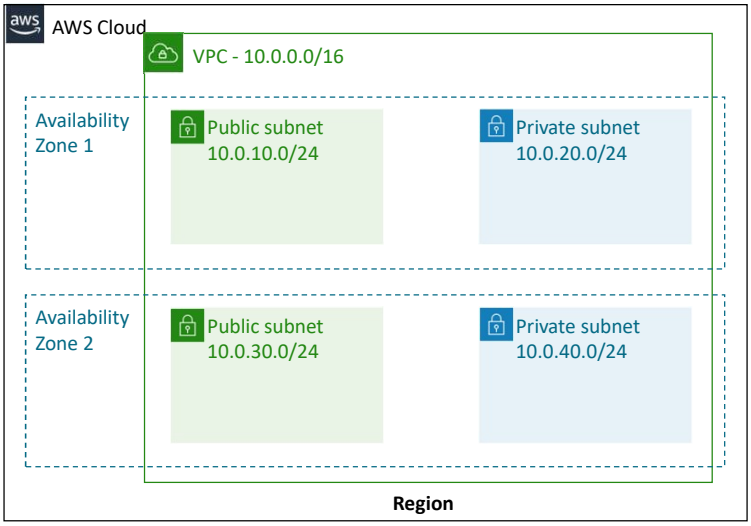
Amazon Virtual Private Cloud (Amazon VPC)

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Subnets

- **Public subnet**
  - The subnet has a direct route to an internet gateway.
- **Private subnet**
  - The subnet does not have a direct route to an internet gateway.
  - Resources in a private subnet require a NAT device to access the public internet.

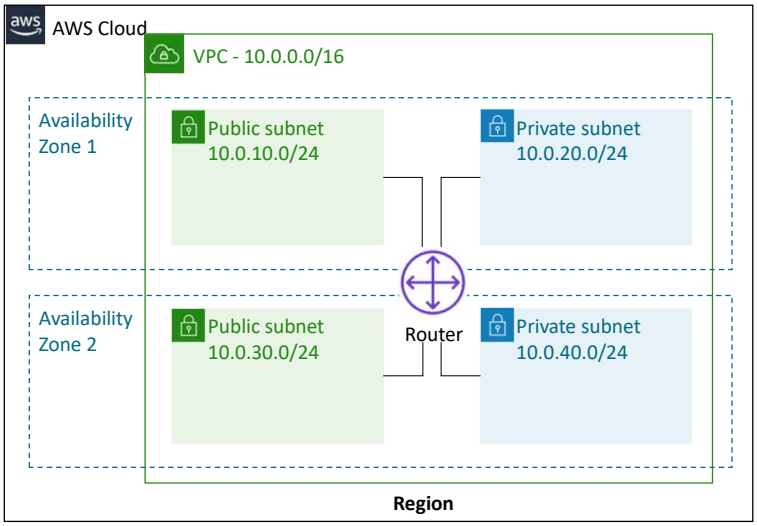


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Route table

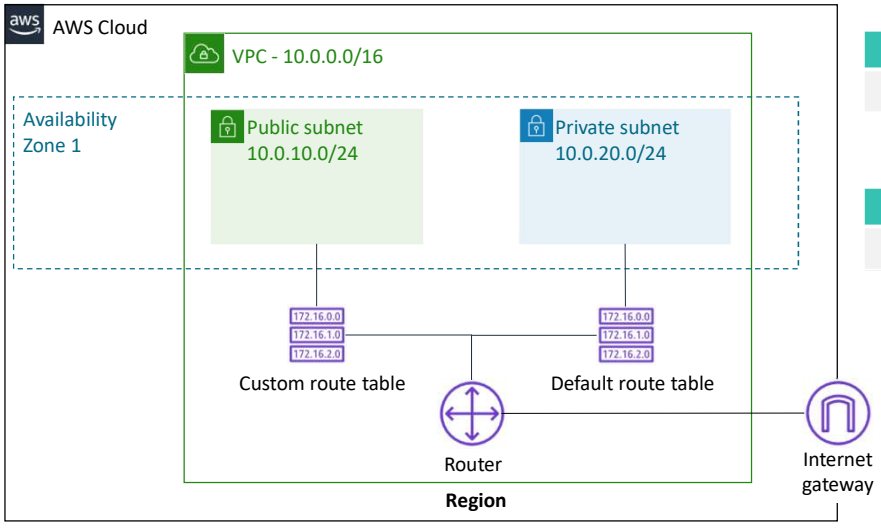
- **Main route table**
  - Automatically created.
  - When a subnet does not have an explicit routing table associated with it, the main routing table is used by default.
- **Custom route table**
  - A route table that you create for your VPC.



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Internet gateway



Default Route Table

Destination	Target
10.0.0.0/16	local

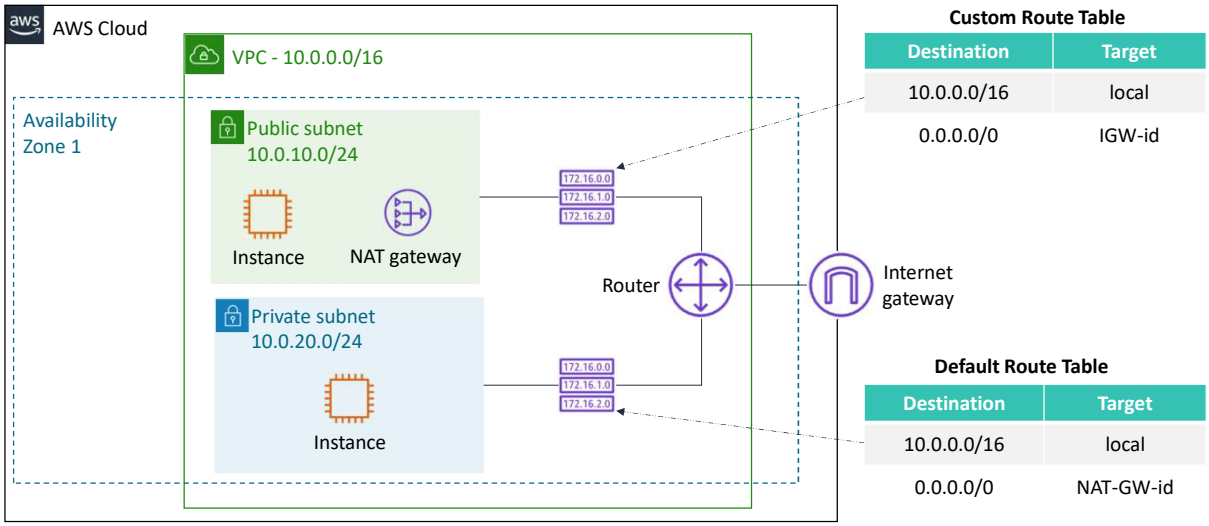
Custom Route Table

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	IGW-id

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NAT gateway

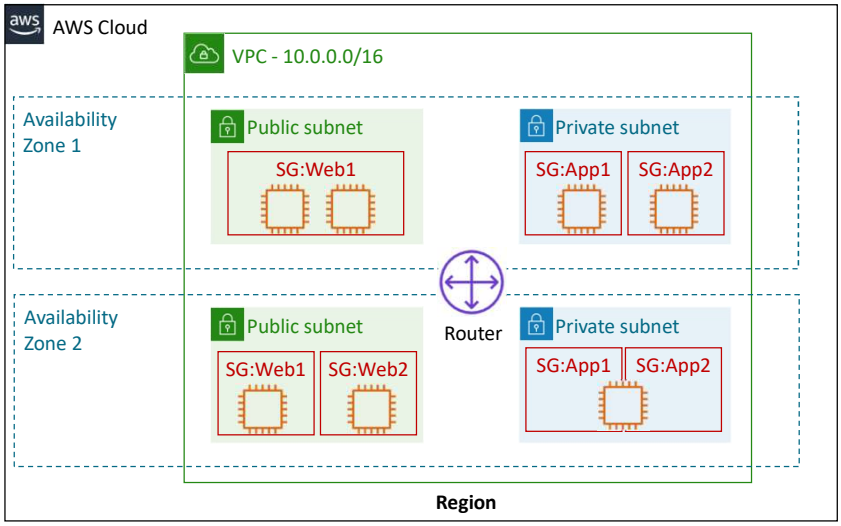


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Security groups



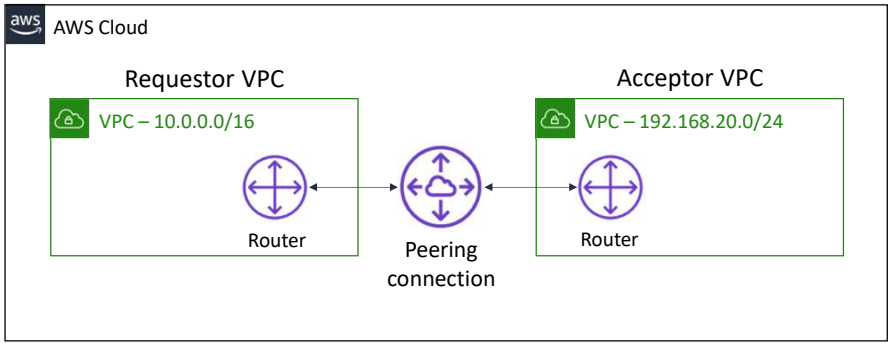
- Instance/elastic network interface layer
- Controls traffic in and out
- Uses Allow rules
- Defaults:
  - Deny all inbound
  - Allow all outbound
- Stateful

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VPC peering



Route Table

Destination	Target
10.0.0.0/16	local
192.168.20.0/24	*VPC-peering-id*

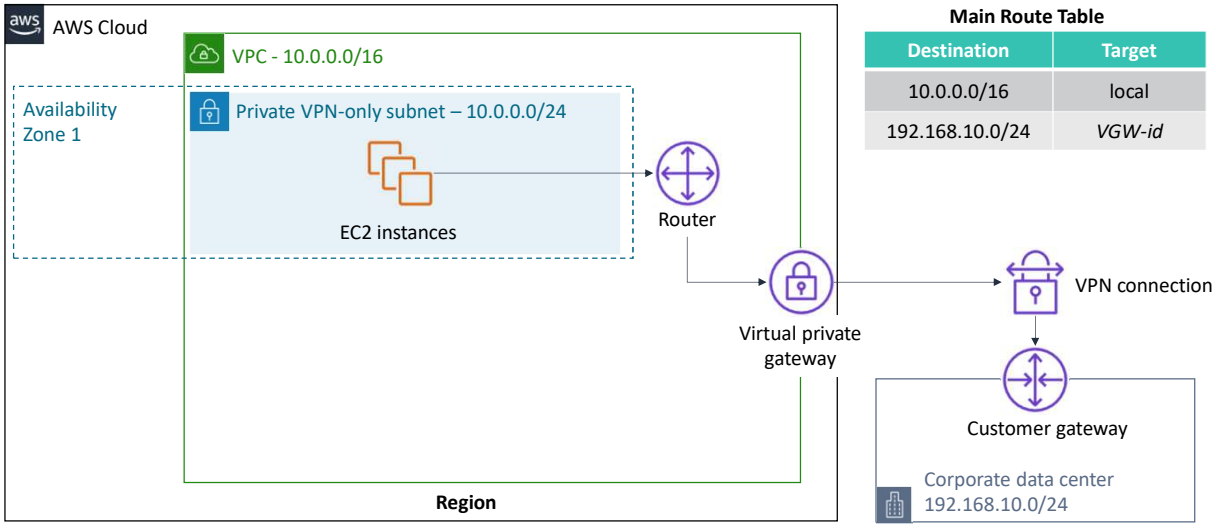
Route Table

Destination	Target
192.168.20.0/24	local
10.0.0.0/16	*VPC-peering-id*

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VPN connections to your data center



Main Route Table

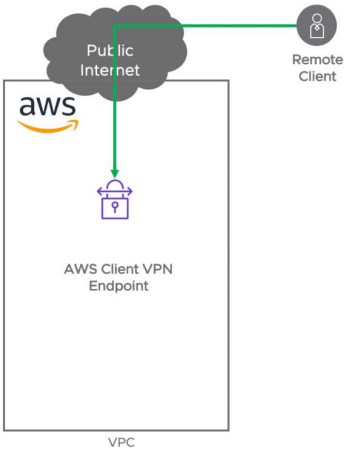
Destination	Target
10.0.0.0/16	local
192.168.10.0/24	*VGW-id*

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Client VPN

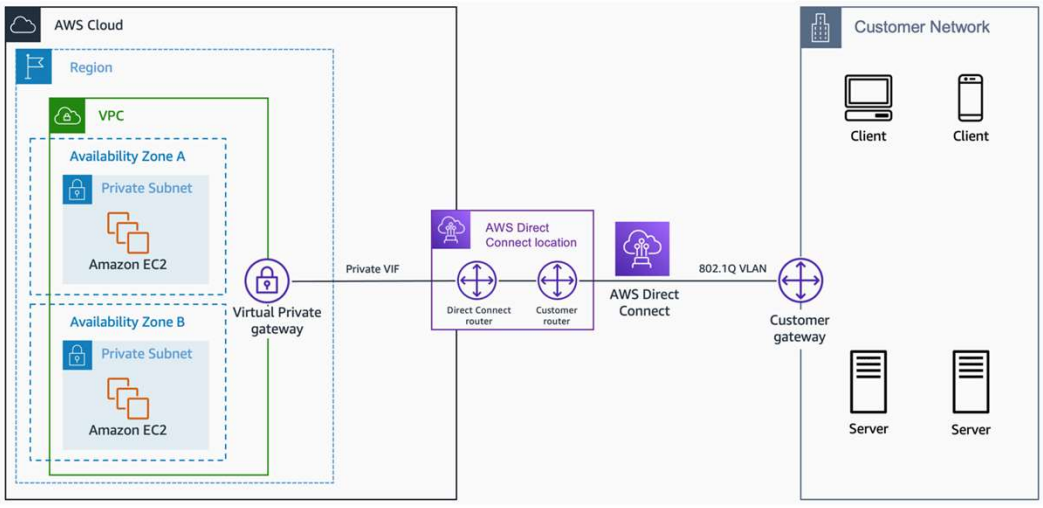
- AWS Client VPN is a managed client-based VPN service.
- Using AWS Client VPN, you can securely access AWS resources and resources in your on-premises network.
- The client establishes the VPN session from their local computer or mobile device using an VPN client application.



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AWS Direct Connect



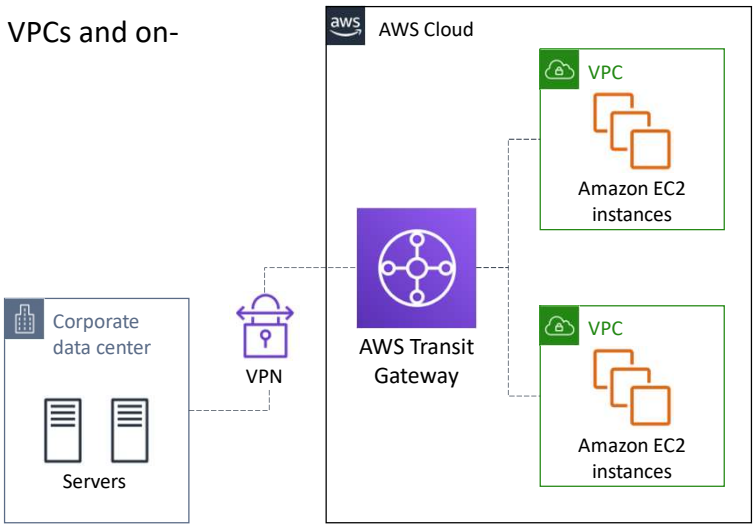
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AWS Transit Gateway

Single gateway to connect multiple VPCs and on-premises networks

- Attach to VPCs and VPNs.
- Propagate routes from the route tables for the attached VPCs and VPN connections.

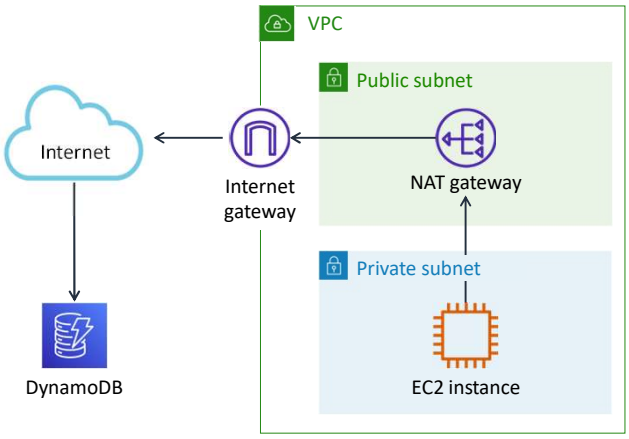


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Without a VPC endpoint

- VPC-based resources must access serverless service URLs through the public internet.



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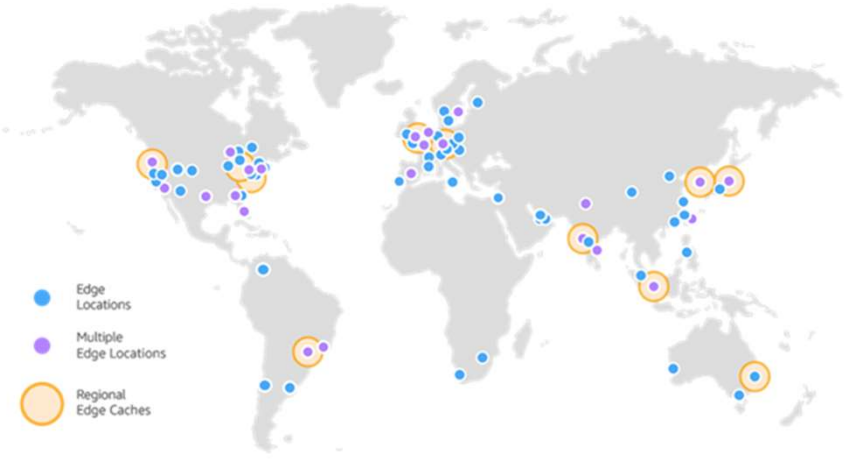
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CloudFront

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Amazon CloudFront infrastructure



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Elastic Load Balancing

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Types of load balancers

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Application Load Balancer	Network Load Balancer	Gateway Load Balancer
HTTP HTTPS	TCP UDP	IP
• Flexible application management • Advanced load balancing of HTTP and HTTPS traffic • Operates at the request level (Layer 7)	• Extreme performance and static IP for your application • Load balancing of TCP traffic • Operates at the connection level (Layer 4)	• Third-party virtual appliance management • Load balancing of traffic through virtual networking appliances • Includes Layer 3 Gateway and Layer 4 load balancer capabilities

Classic Load Balancer	
PREVIOUS GENERATION for HTTP, HTTPS, and TCP	
• For applications that use the EC2-Classic network	• Operates at both the request level and the connection level

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Amazon EC2 Auto Scaling

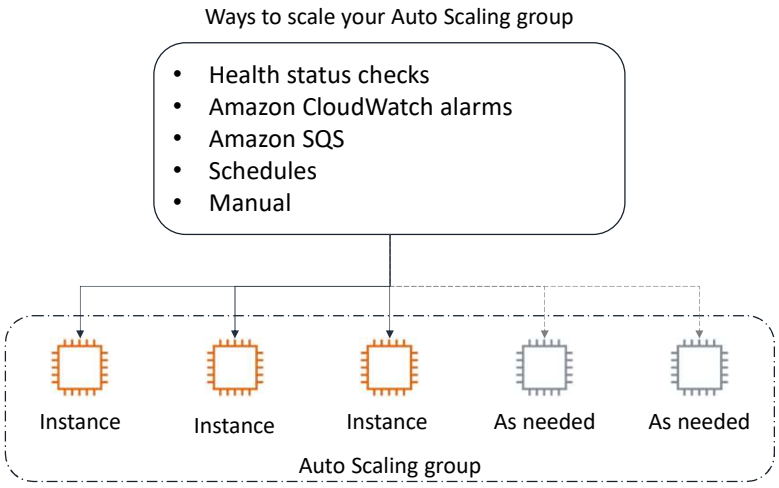
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What is EC2 Auto Scaling?



Amazon EC2 Auto Scaling

Scale out to meet demand.
Scale in to reduce costs.



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Amazon Route 53

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Amazon Route 53



Distributes traffic across *AWS Regions*

A scalable Domain Name System (DNS) web service

- Resolves domain names to IP addresses
- Routes requests based on latency, health checks, and other criteria
- Registers or transfers a domain name

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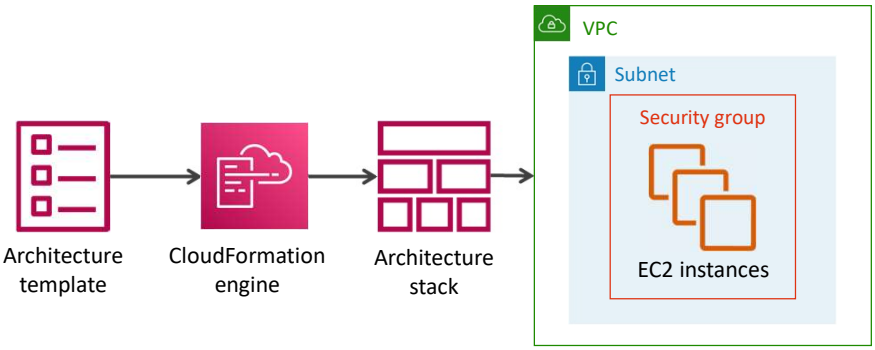
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AWS CloudFormation

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Infrastructure as code (IaC)

- Replicate, redeploy, and repurpose
- Control versioning on infrastructure and applications
- Detect drift
- Roll back the service to the last good state



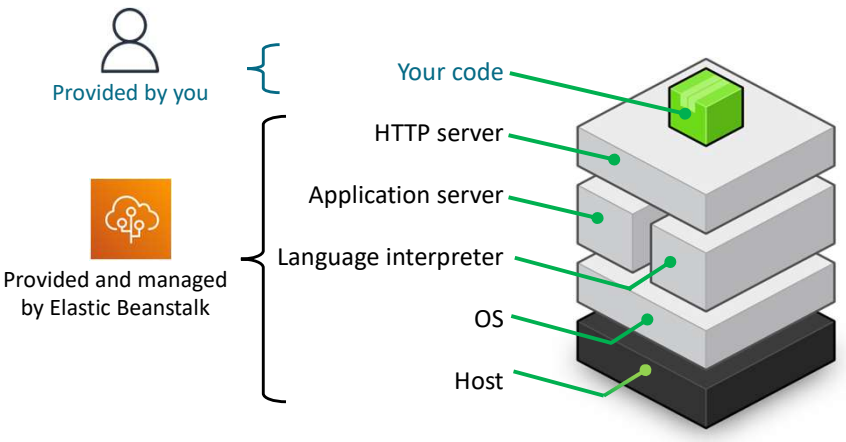
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AWS Elastic Beanstalk

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AWS Elastic Beanstalk

- Provisions and operates the infrastructure
- Manages the application stack for you
- Shows everything that is created
- Automatically scales your application up and down

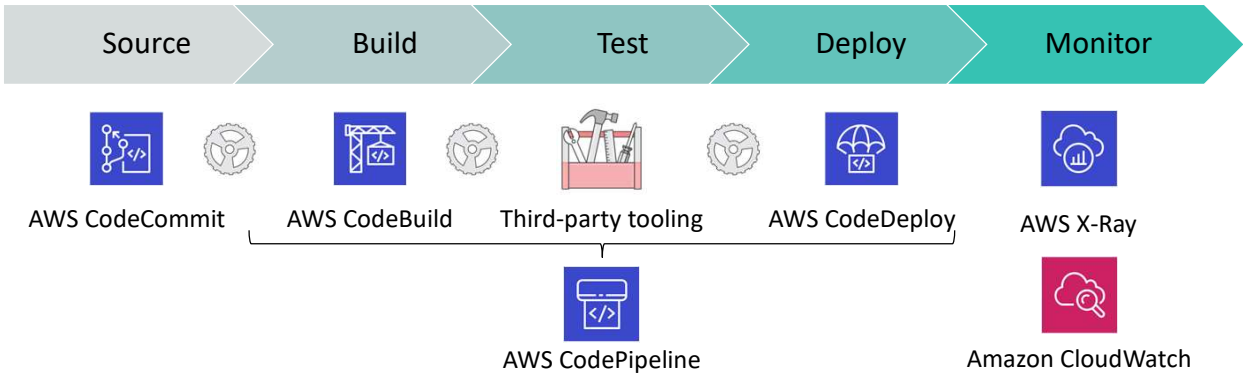


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DevOps

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AWS code services



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Amazon Elastic Block Store (Amazon EBS)

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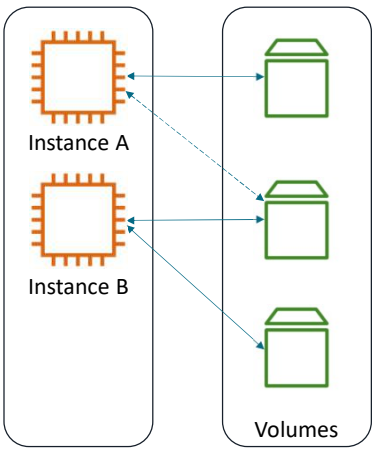
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Amazon EBS



Amazon EBS

- Network-attached
- Persistent block storage
- Boot and data volumes supported
- Multi-Attach



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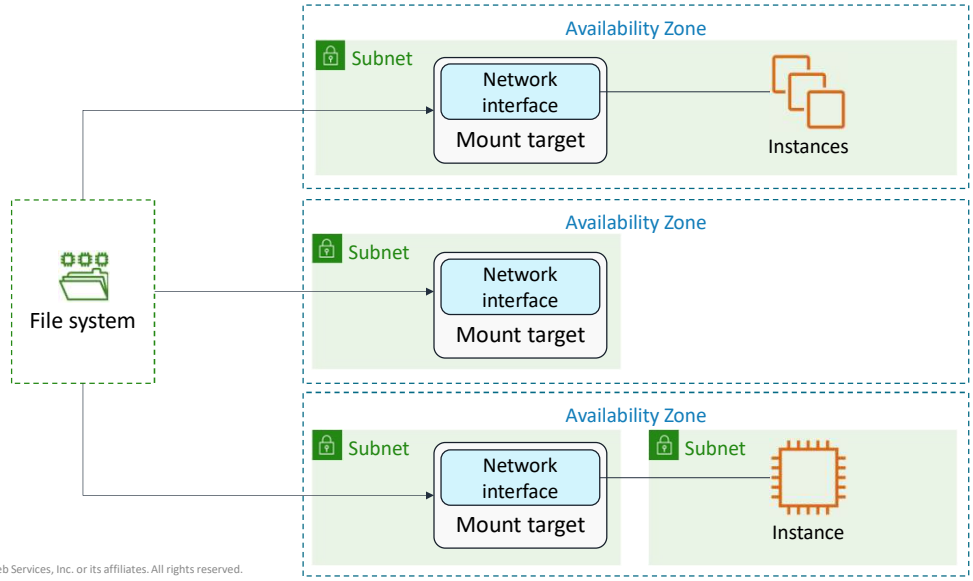
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Shared file system storage

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Amazon Elastic File System (Amazon EFS)



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Amazon Simple Storage Service (Amazon S3)

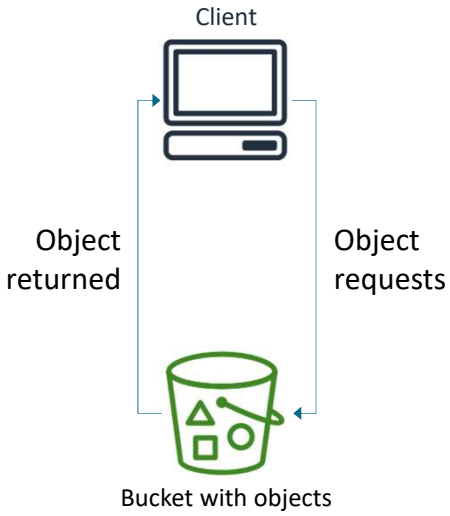
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Amazon S3



Amazon S3

- Fast, durable, highly available key-based access to objects
- Object storage built to store and retrieve data
- Not a file system



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Object storage classes

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S3 Standard


S3 Intelligent-Tiering


S3 Standard-IA


S3 One Zone-IA


S3 Glacier


S3 Glacier Deep Archive

Access frequency: Frequent ⟷ Infrequent

	S3 Standard	S3 Intelligent-Tiering	S3 Standard-IA	S3 One Zone-IA	S3 Glacier	S3 Glacier Deep Archive
Availability Zones	≥3	≥3	≥3	1	≥3	≥3
Monitoring fees		Per object				
Retrieval fees			Per GB	Per GB	Per GB	Per GB

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AWS Backup

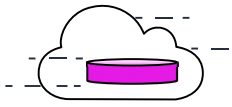
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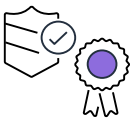
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AWS Backup

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Cloud-native Backups



Compliance



Disaster Recovery



Policy- and tag-based backup solution



Automated backup scheduling



Centralized backup activity monitoring and logs



Backup access policies



Backup encryption



Automated backup retention management



No added cost for orchestration

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RDS

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Amazon RDS engines

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Commercial	Open source	Cloud native
Microsoft SQL Server	PostgreSQL	Amazon Aurora
ORACLE	MySQL	PostgreSQL
	MariaDB	MySQL

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AWS DynamoDB

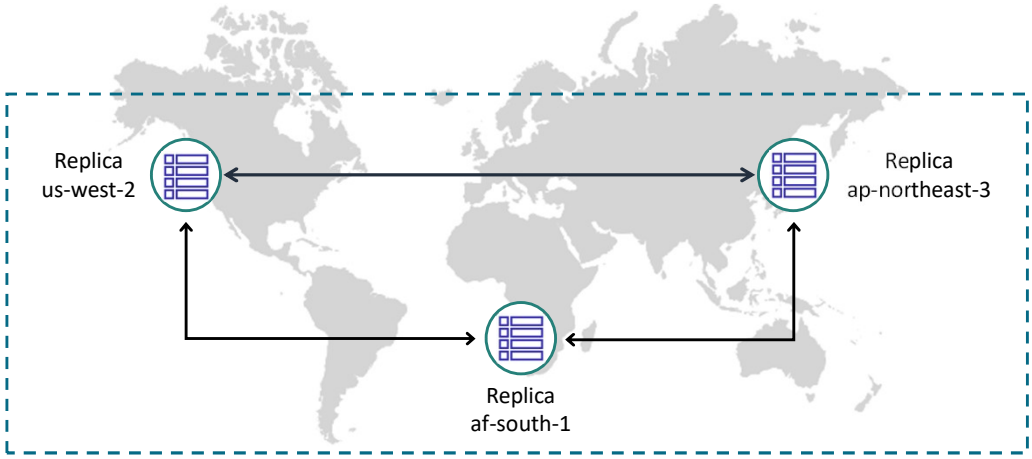
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DynamoDB global tables

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- Global tables automate replication across Regions



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Data Security

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AWS Key Management Service



AWS KMS

- AWS Key Management Service (KMS) gives you centralized control over the cryptographic keys used to protect your data.
- You can use a KMS key to encrypt, decrypt, and re-encrypt data.
- An AWS KMS key is a logical representation of a cryptographic key.
- AWS KMS contains a reference to the key material* used to perform cryptographic operations.

*Key material is string of bits used in a cryptographic algorithm.

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AWS Certificate Manager



AWS Certificate Manager

- AWS Certificate Manager (ACM) simplifies the provisioning, managing, and deploying, SSL/TLS certificates for use with AWS services.
- Removes the manual process of purchasing, uploading, and renewing certificates.
- Requests and provisions certificates and deploys them for sites that use ELB or CloudFront.
- Avoids downtime due to misconfigured, revoked, or expired certificates by managing renewals automatically.

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AWS Secrets Manager



AWS Secrets Manager

- Secret management allows users to safely and securely store secrets in a central repository.
- AWS Secrets Manager helps you manage, retrieve, and rotate database credentials, application credentials, OAuth tokens, API keys, and other secrets throughout their lifecycles.
- With proper access controls, only specific entities can retrieve the decrypted values of these secrets.

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Amazon GuardDuty

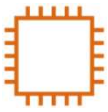


Amazon GuardDuty

GuardDuty is a threat detection service that continuously monitors for malicious activity and unauthorized behavior.



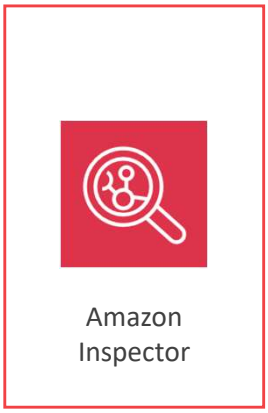
AWS accounts



Workloads

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Amazon Inspector



- Amazon Inspector is a centrally manage vulnerability management service.
- Amazon Inspector scans, running Amazon EC2 instances, container images in Amazon ECR, and AWS Lambda functions.
- Amazon Inspector scans for software vulnerabilities and unintended network exposure and creates a finding.
- Findings include, details associated with the detection to remediate the vulnerability and provides remediation guidance.

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AWS Shield

- Managed DDoS protection service
- AWS Shield Standard at no charge
- AWS Shield Advanced provides integration with AWS WAF and protection of CloudFront distributions



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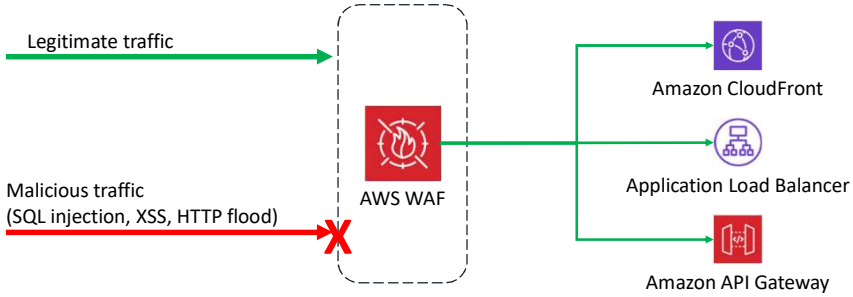
AWS WAF

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AWS WAF

Helps protect your web applications and APIs against common web exploits



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AWS CloudWatch

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CloudWatch benefits

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Amazon CloudWatch

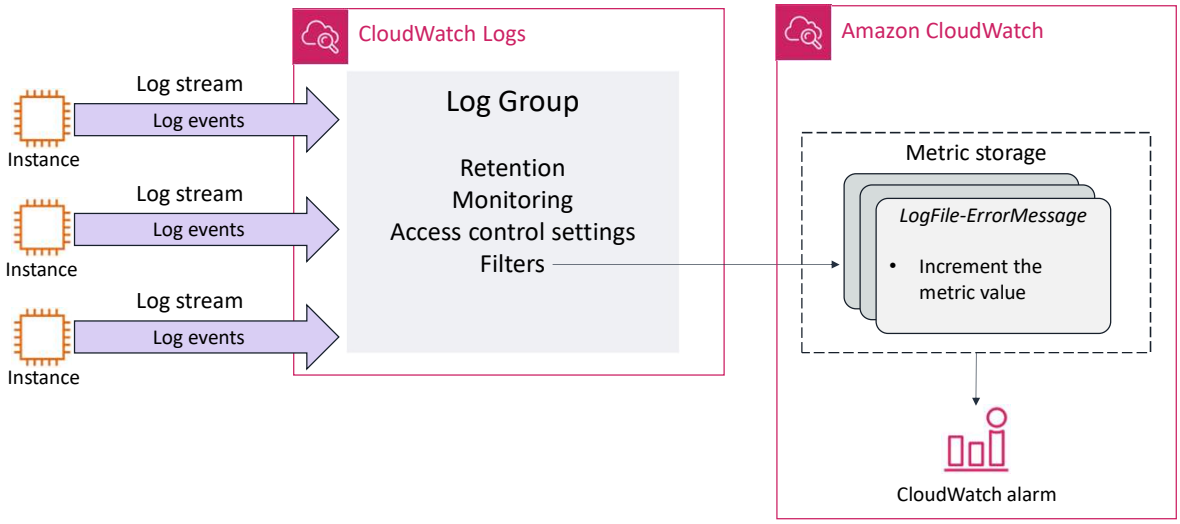
- Central access to all data
- Simple initiation of collecting metrics
- Comprehensive exploration, analysis, and visualization of logs
- Automated responses
- Real-time and historical data

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CloudWatch Logs functionality

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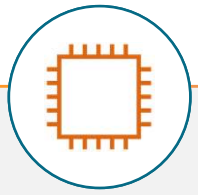
CloudTrail

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Examples of CloudTrail logged activity

Log, monitor, and retain account activity related to actions across your AWS infrastructure.



Who shut down a specific instance?



Who changed a security group configuration?



What activities were denied due to lack of permissions?

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Migration Tools and Services

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Moving large data into Amazon S3



AWS Snowcone

- Snowcone is a small, rugged, edge computing and data storage product.



AWS Snowball Edge

- Snowball Edge is an edge computing and data transfer device.

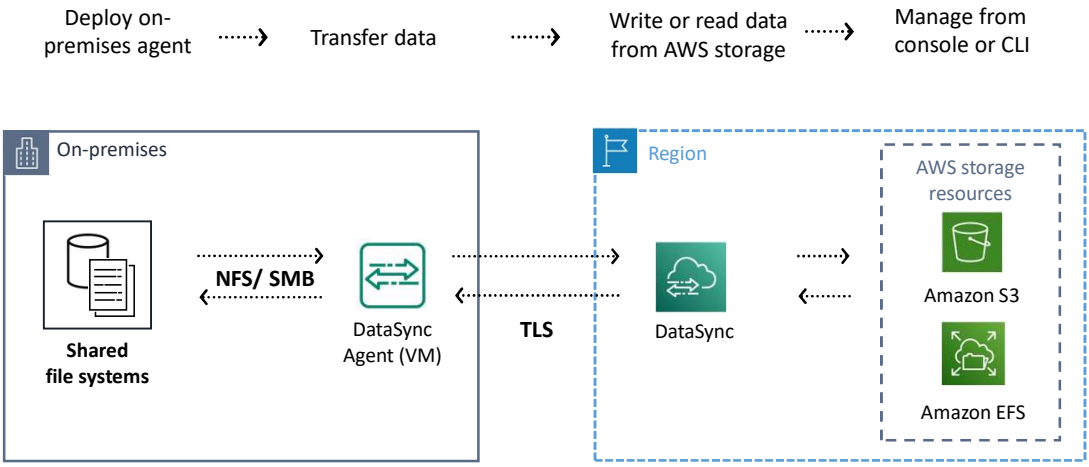


AWS Snowmobile

- Snowmobile is the first exabyte-scale data migration service to move very large datasets from on premises to AWS.

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DataSync



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AWS Transfer Family

- AWS Transfer Family is a secure transfer service that enables you to transfer files into and out of AWS storage services.
- AWS Transfer Family supports transferring data from or to the following:
  - Amazon Simple Storage Service (S3) storage.
  - Amazon Elastic File System (EFS) Network File System (NFS) file systems.
- AWS Transfer Family supports transferring data over the following protocols:
  - Secure Shell (SSH) File Transfer Protocol (SFTP)
  - File Transfer Protocol Secure (FTPS)
  - File Transfer Protocol (FTP)

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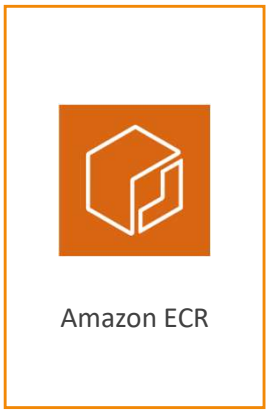
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Amazon Elastic Container Registry (ECR)

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Amazon Elastic Container Registry



- Fully managed AWS Docker registry service.
- Supports private repositories with resource-based permissions using AWS IAM.
- You can use your preferred CLI to push, pull, and manage Docker images and compatible artifacts.

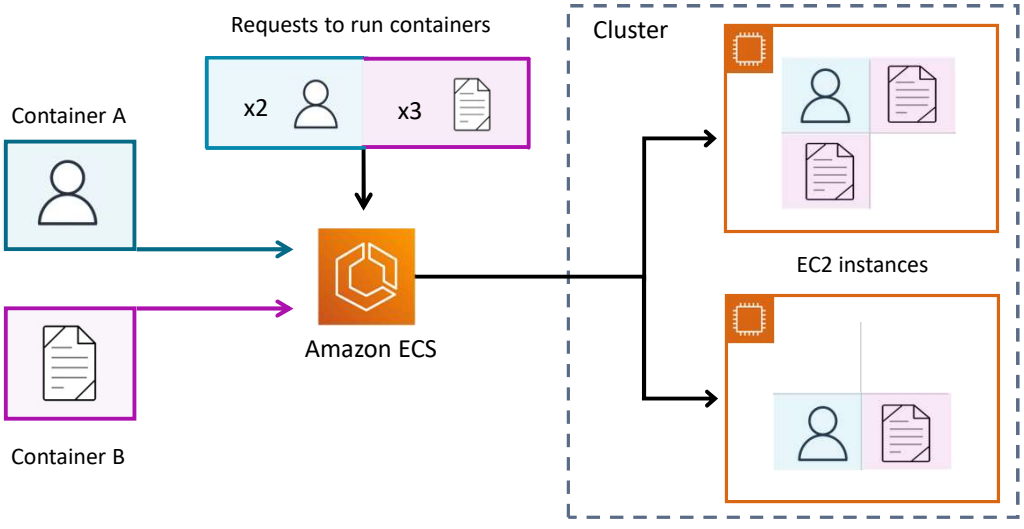
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Amazon Elastic Container Service (ECS)

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Amazon ECS orchestration



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Amazon Elastic Kubernetes Service (EKS)

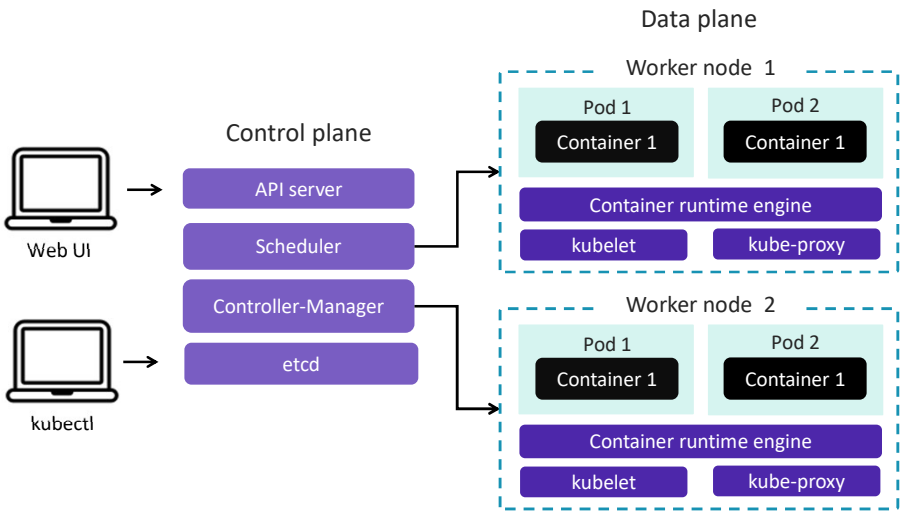
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Kubernetes architecture

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- A Kubernetes cluster, is a set of worker machines called nodes, that run containerized applications.



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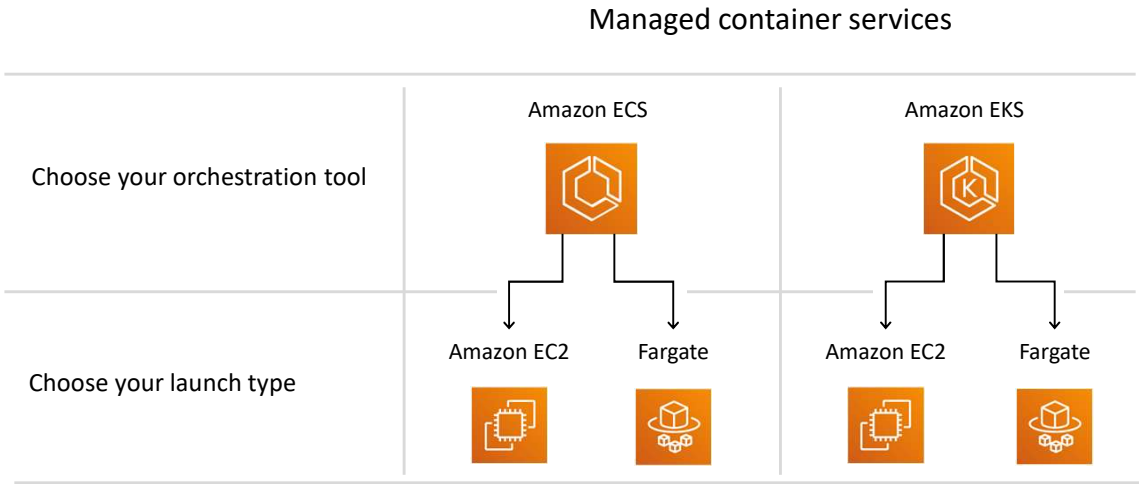
Amazon Fargate

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Amazon ECS and Amazon EKS

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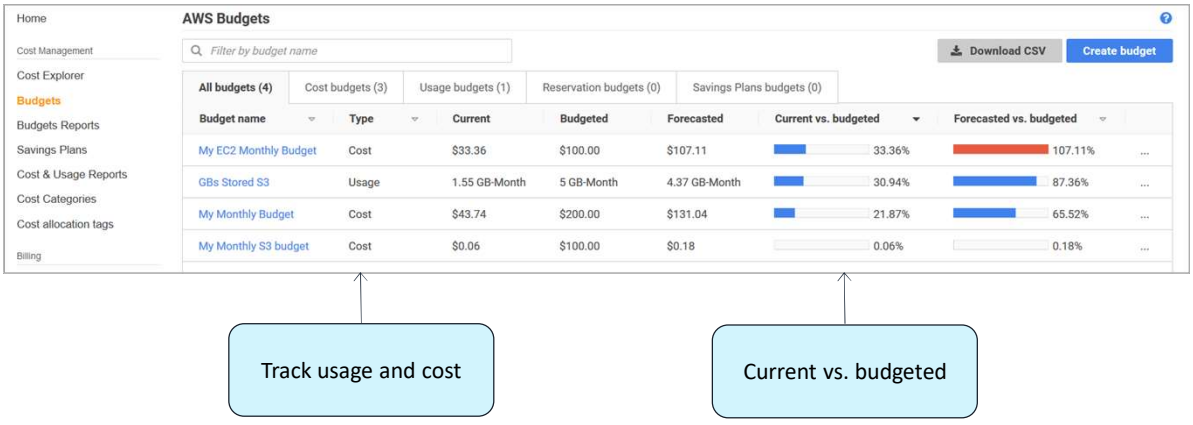
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AWS Budget

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AWS Budgets



Track usage and cost

Current vs. budgeted

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AWS Trusted Advisor

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Trusted Advisor

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Each category will provide suggestions and show warnings for high priority items



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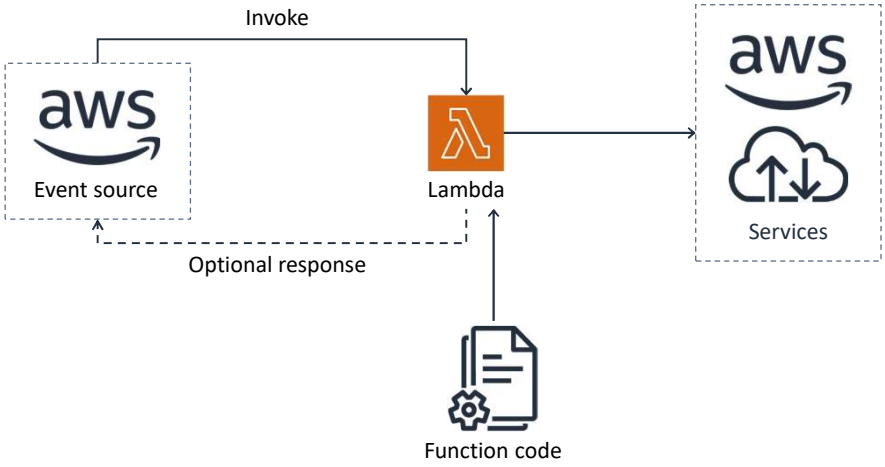
Lambda

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Lambda

- Serverless compute
- Supports Node.js, Java, Python, C#, Go, PowerShell, Ruby, and more
- Runs for up to 15 minutes
- Supports up to 10 GB memory



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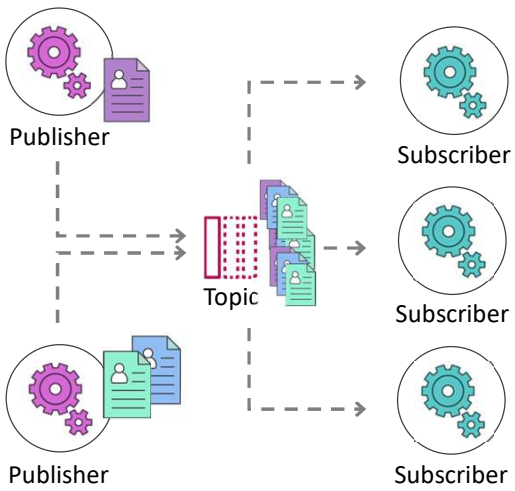
Amazon Simple Notification Service

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Amazon Simple Notification Service

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Types of subscribers

- Email/Email-JSON
- Mobile text messaging (SMS)
- Mobile push notification
- HTTP/HTTPS
- AWS Lambda
- Amazon SQS
- Kinesis Data Firehose

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Simple Queue Service

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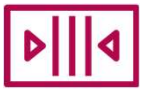
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Amazon Simple Queue Service

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Amazon Simple Queue Service (Amazon SQS)



Fully managed message queueing service



Stores messages until they are processed and deleted



Acts as a buffer between senders and receivers

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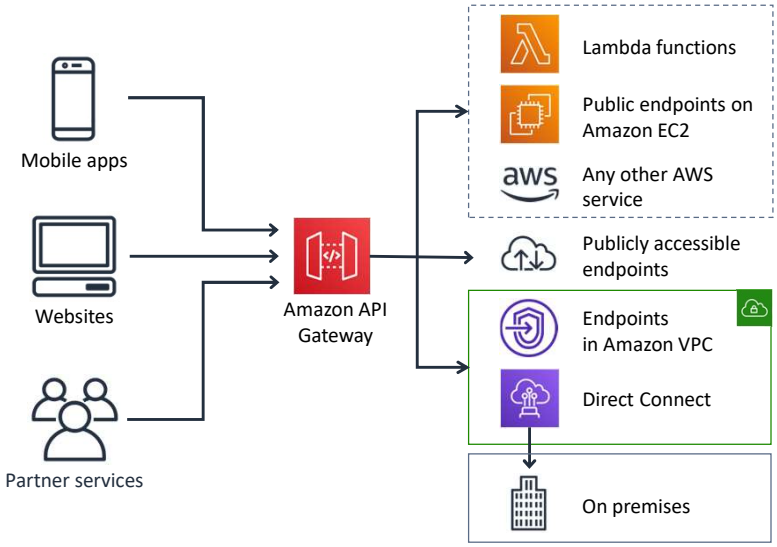
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Amazon API Gateway

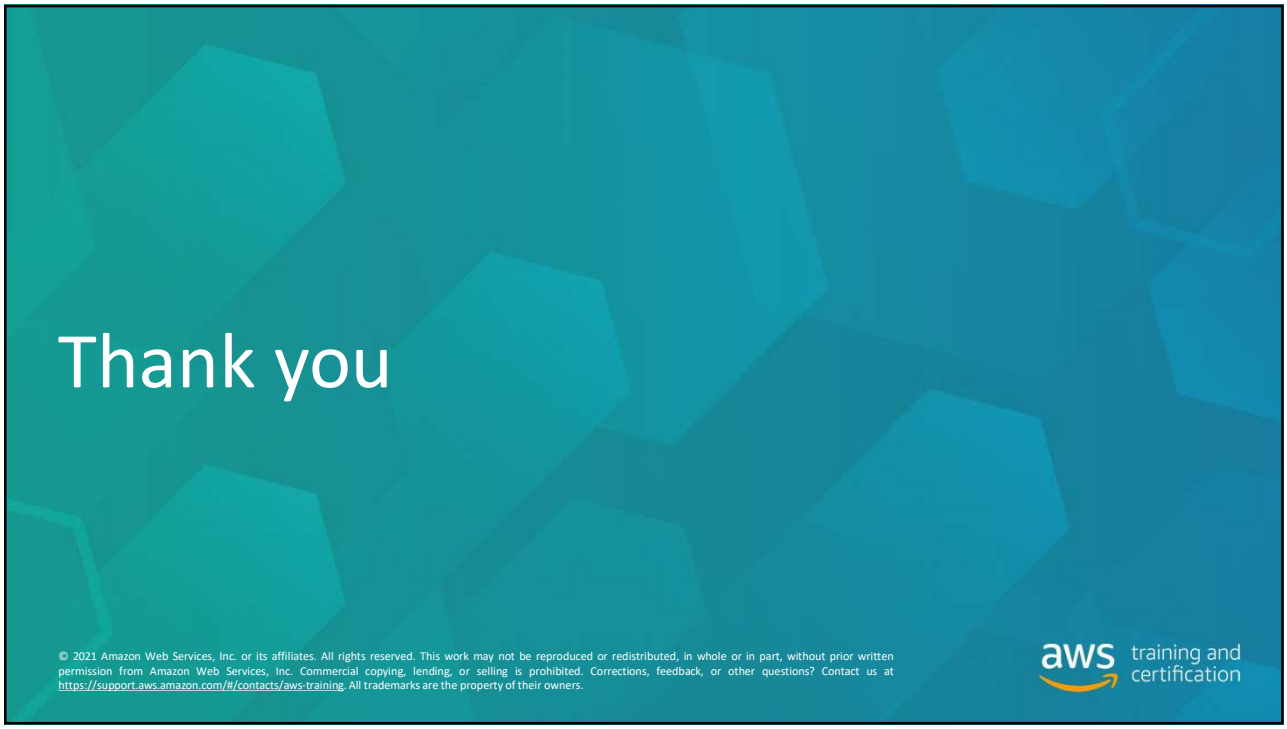
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Amazon API Gateway

- Create an entry point for your applications.
- Process thousands of concurrent API calls.
- Choose internet facing or internal only.



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